Overcoming catastrophic forgetting in neural networks

Kirkpatrick et al. (2017)

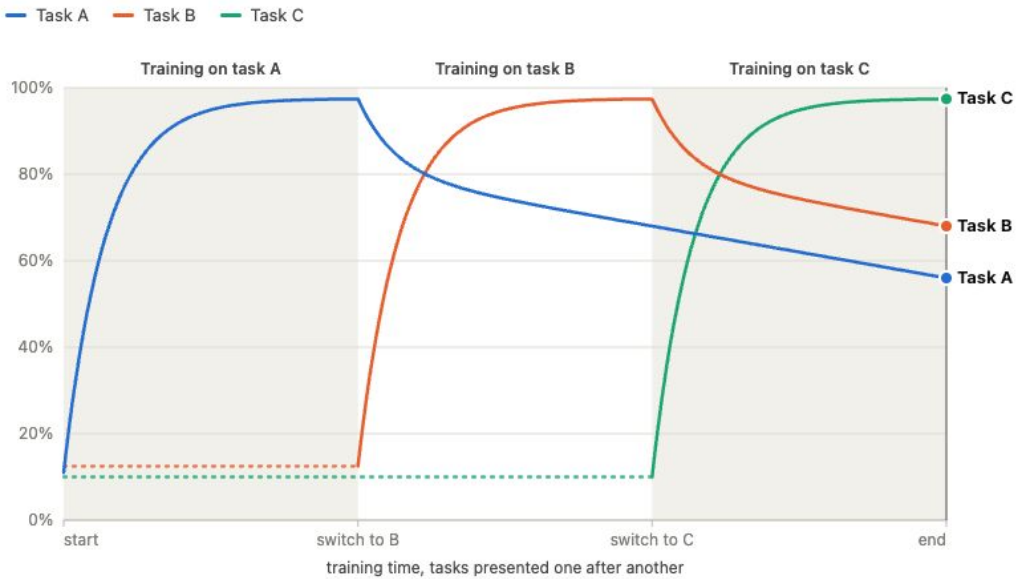
Presented by Marius Mosbach

Motivation

- General intelligence requires agents that can learn and remember many tasks
- Agents need to demonstrate *continual learning*
- The ability to learn consecutive tasks without forgetting to perform previous ones

Catastrophic forgetting

- Occurs specifically when models are trained sequentially on multiple tasks
- Common solution so far: interleaving data



Continual learning in humans and animals

- Humans and other animals appear to be able to learn continually
- How do they avoid catastrophic forgetting?
- Recent* evidence: acquired knowledge is protected in neocortical circuits
- Causal evidence that *protection* of certain synapses is critical for retention 🐁
- This provides a biology inspired motivation for the work

* Keep in mind this paper was written in 2017

Main contributions

1. A new algorithm for continual learning (EWC)
    a. Specifically, learning N consecutive tasks without forgetting
2. Experimental verification in two settings
    a. Supervised classification (MNIST)
    b. RL (Atari games)

Remainder of this presentation

1. Elastic weight consolidation (EWC)
2. Supervised learning results
3. RL results
4. Discussion and summary

Elastic Weight Consolidation (EWC)

- Assume we found a good solution for task A already
- There are many possible solutions that achieve low error on task B
- We want to find solutions that give low error for task A **and** task B

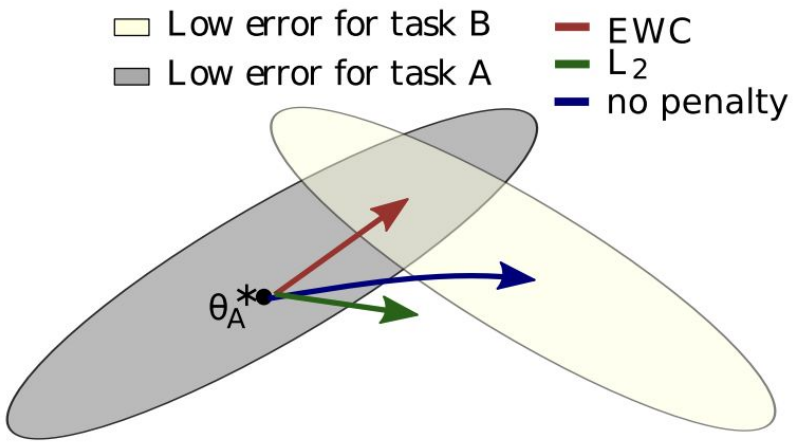


Figure 1 by Kirkpatrick et al. (2017)

How do we find such a solution?

Elastic Weight Consolidation (EWC)

- Take a Bayesian perspective
- Assume we found a local minimum for task A
- We are interested in the shape of the weight space around it

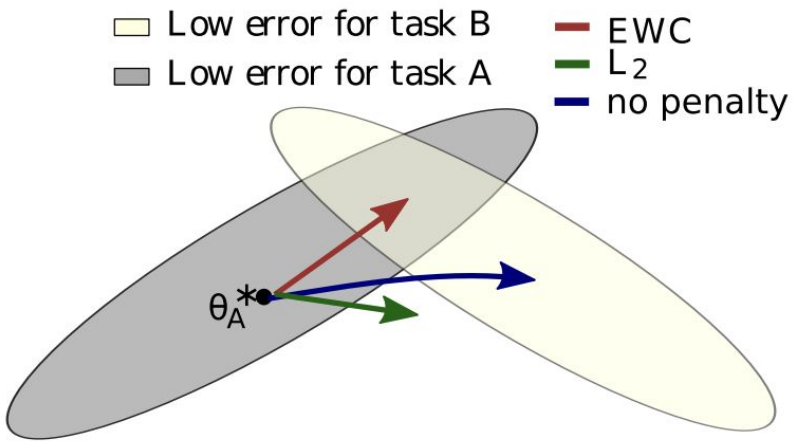


Figure 1 by Kirkpatrick et al. (2017)

Detour: Bayesian view on training

Likelihood

$p(\mathcal{D} \mid \theta)$

How well do the weights explain the data?

Prior

$p(\theta)$

What do we believe before seeing data?

Posterior

$p(\theta \mid \mathcal{D})$

What do we believe after seeing data?

Note that D is a dataset and theta are the weights/parameters of the model

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Likelihood: how probable is the data under the model?

$$p(\mathcal{D} \mid \theta) = \prod_{n=1}^{N} p(y_n \mid x_n, \theta)$$

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Likelihood: how probable is the data under the model?

$$\log p(\mathcal{D} \mid \theta) = \sum_{n=1}^{N} \log p(y_n \mid x_n, \theta)$$

Bayes' rule: updating beliefs with data

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}$$

Posterior | Likelihood | Prior | Evidence (constant in θ)

Maximum likelihood vs. maximum a posteriori

$$\theta_{\mathrm{ML}} = \arg\max_{\theta} \; p(\mathcal{D} \mid \theta)$$

Both return a single point

$$\theta_{\mathrm{MAP}} = \arg\max_{\theta} \; p(\theta \mid \mathcal{D}) = \arg\min_{\theta} \Big[\mathcal{L}(\theta) \; - \; \log p(\theta) \Big]$$

Bayes' rule

$$p(\theta \mid \mathcal{D}) \; = \; \frac{p(\mathcal{D} \mid \theta) \; p(\theta)}{p(\mathcal{D})}$$

The loss

$$\mathcal{L}(\theta) \; = \; - \log p(\mathcal{D} \mid \theta)$$

Posterior | Likelihood | Prior | Evidence (constant in θ)

Solving the MAP problem

Assume the tasks are independent

$$p(\mathcal{D} \mid \theta) = p(\mathcal{D}_A \mid \theta)\, p(\mathcal{D}_B \mid \theta) \qquad p(\mathcal{D}) = p(\mathcal{D}_A)\, p(\mathcal{D}_B)$$

Posterior Likelihood Prior Evidence (constant in θ)

Solving the MAP problem

Assume the tasks are independent

$$p(\mathcal{D} \mid \theta) = p(\mathcal{D}_A \mid \theta)\, p(\mathcal{D}_B \mid \theta) \qquad p(\mathcal{D}) = p(\mathcal{D}_A)\, p(\mathcal{D}_B)$$

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D}_A \mid \theta) + \log p(\mathcal{D}_B \mid \theta) + \log p(\theta) - \log p(\mathcal{D}_A) - \log p(\mathcal{D}_B)$$

Bayes' rule and taking the log

Posterior | Likelihood | Prior | Evidence (constant in θ)

Solving the MAP problem

Assume the tasks are independent

$$p(\mathcal{D} \mid \theta) = p(\mathcal{D}_A \mid \theta)\, p(\mathcal{D}_B \mid \theta) \qquad p(\mathcal{D}) = p(\mathcal{D}_A)\, p(\mathcal{D}_B)$$

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D}_A \mid \theta) + \log p(\mathcal{D}_B \mid \theta) + \log p(\theta) - \log p(\mathcal{D}_A) - \log p(\mathcal{D}_B)$$

Regroup terms that belong to task A only

$$\underbrace{\log p(\mathcal{D}_A \mid \theta) + \log p(\theta) - \log p(\mathcal{D}_A)}_{= \log\ p(\theta|\mathcal{D}_A)\ \text{(Bayes' rule on } A)}$$

Posterior | Likelihood | Prior | Evidence (constant in θ)

Equation (2) from the paper

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D}_B \mid \theta) + \log p(\theta \mid \mathcal{D}_A) - \log p(\mathcal{D}_B)$$

- The ordinary loss (likelihood) on data from B
- Tasks A's posterior now acting as the prior, i.e., everything we learned from A
- A constant term that does not depend on the weights

Posterior | Likelihood | Prior | Evidence (constant in θ)

Note: The posterior is ~ the loss surface

$$-\log p(\theta \mid \mathcal{D}_A) = \mathcal{L}_A(\theta) - \log p(\theta) + \text{const}$$

- Computing the posterior exactly is intractable
- So we need to approximate it near the solution we found for task A
- Importantly: we can do this only once because we won't have data from A anymore

Posterior | Likelihood | Prior | Evidence (constant in θ)

Approximating the posterior

$$H = -\nabla^2 \log p(\theta \mid \mathcal{D}_A)\Big|_{\theta=\theta^*_A}$$

$$\log p(\theta \mid \mathcal{D}_A) \approx \text{const} - \frac{1}{2}(\theta - \theta^*_A)^\top H (\theta - \theta^*_A)$$

- We approximate the posterior with a second-order Taylor expansion
- Assumption: our solution for task A is a local minimum
- The Hessian is the curvature of the loss

We assume the posterior is a Gaussian

Posterior | Likelihood | Prior | Evidence (constant in θ)

Approximating the Hessian

$$F_i = \mathbb{E}_{x \sim \mathcal{D}_A} \mathbb{E}_{y \sim p(y|x,\theta_A^*)} \left[\left(\frac{\partial \log p(y \mid x, \theta)}{\partial \theta_i} \right)^2 \right]_{\theta = \theta_A^*}$$

- Note: Inputs are from task A; we use only model predictions, not labels
- Near a minimum, the Fisher is close to the Hessian
  - But needs only first order derivatives

$$\frac{1}{2}(\theta - \theta_A^*)^\top H (\theta - \theta_A^*) \approx \frac{1}{2}(\theta - \theta_A^*)^\top F (\theta - \theta_A^*)$$

Keeping only the diagonal

$$\frac{1}{2}(\theta - \theta^*_A)^\top H\,(\theta - \theta^*_A) \;\approx\; \frac{1}{2}(\theta - \theta^*_A)^\top F\,(\theta - \theta^*_A)$$

$$\frac{1}{2}\,(\theta - \theta^*_A)^\top H\,(\theta - \theta^*_A) \;\approx\; \sum_i \frac{1}{2}\,F_i\,(\theta_i - \theta^*_{A,i})^2$$

- We end up with one number (F_i) per weight
- This is a matrix! N^2 numbers for N weights

Summarizing our simplifications

Step	What we replace	What we gain	What we give up
Laplace	Posterior → Gaussian at θ_A^*	A closed-form quadratic	Valid only near θ_A^*
Fisher	Hessian → Fisher	First derivatives, never negative	Equals the Hessian only near a minimum
Diagonal	Full matrix → one number per weight	Linear cost	Correlations between weights

Let's look at a demo

Elastic Weight Consolidation (EWC)

We arrive at the final objective from their paper by negating equation (2) and plugging in our posterior approximation

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D}_B \mid \theta) + \log p(\theta \mid \mathcal{D}_A) - \log p(\mathcal{D}_B) \tag{2}$$

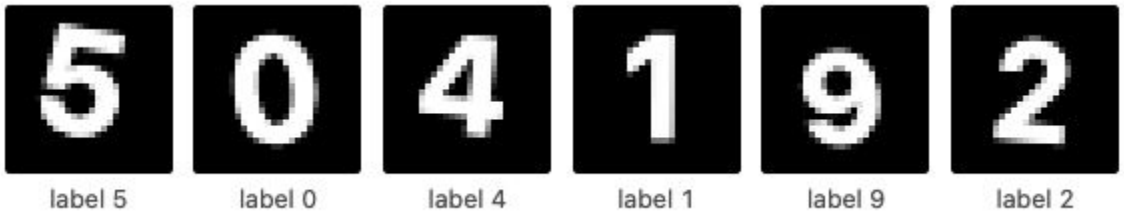
$$-\log p(\theta \mid \mathcal{D}) \simeq \mathcal{L}_B(\theta) + \sum_i \frac{1}{2} F_i \left(\theta_i - \theta^*_{A,i}\right)^2 + \text{const} \tag{3}$$

Optimize only on data from task B!

Posterior | Likelihood | Prior | Evidence (constant in θ)

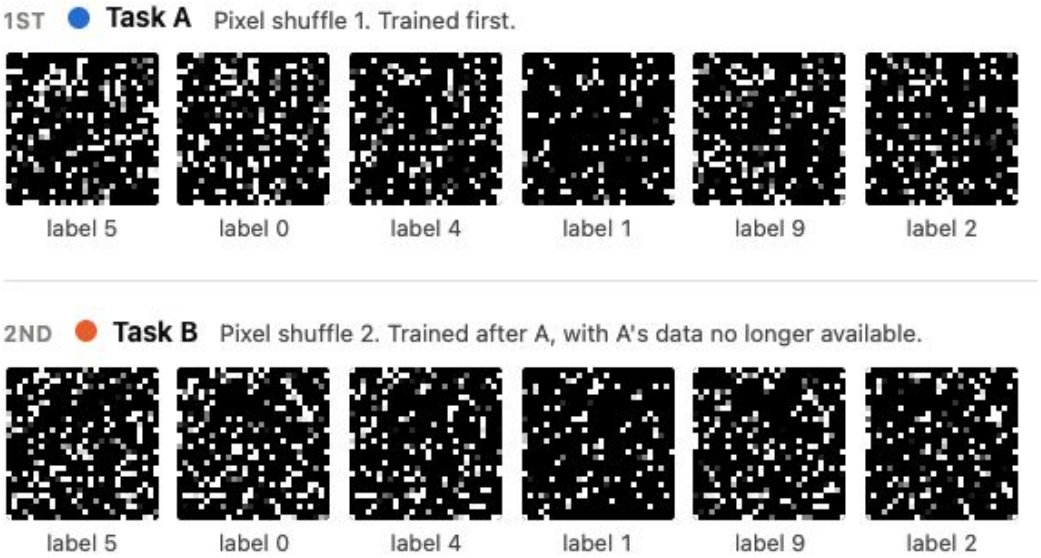
Experiments 1: Supervised classification

- Fully connected MLP
- MNIST dataset, i.e., digit classification



Experiments 1: Supervised classification

- Create different training sets by permuting pixels randomly
- train A, train B, and train C differ based on the random seed used

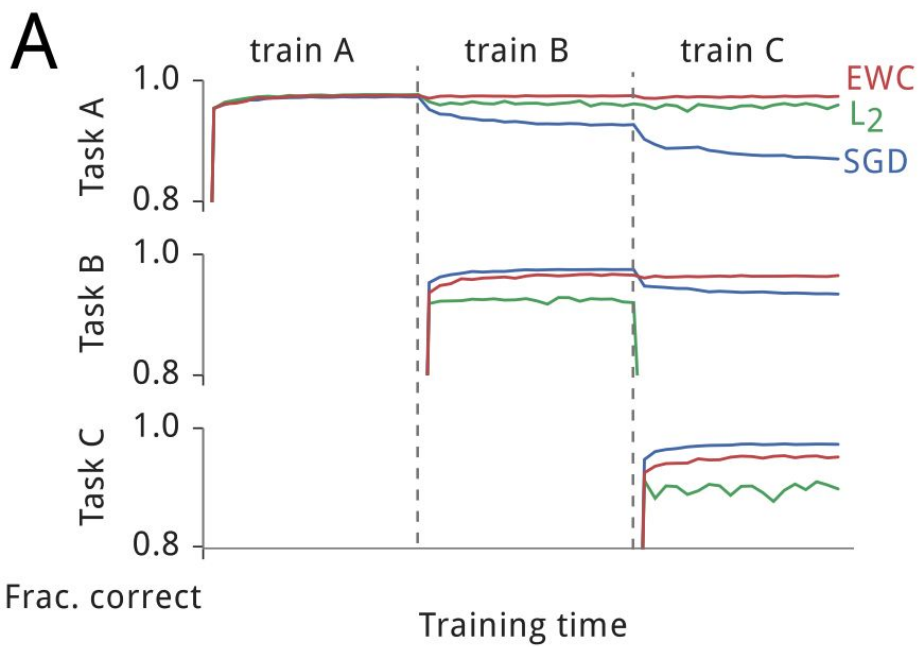


Experiments 1: Supervised classification

- Create different training sets by permuting pixels randomly
- Training sets A, B, and C differ based on the random seed used
- Train sequentially on A, B, C using SGD, SGD+L2 regularization, EWC

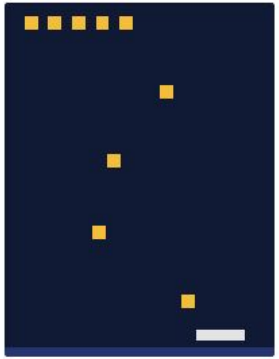
Results

- SGD and SGD+L2 are not sufficient to prevent catastrophic forgetting
- EWC maintains performance on previous tasks

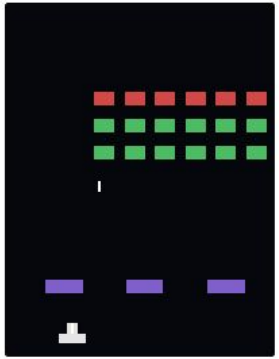


Experiments 2: RL

- DeepQ Networks (DQNs)
- Atari 2600 task set
- 10 games chosen randomly, presented randomly
    - Same game might be seen several times



Game 1: Catch
Slide a bar sideways to catch falling blocks.



Game 2: Shooter
Move a cannon and fire at rows of descending targets.

Experiments 2: RL

- Two important modifications compared to prior work
  - Task-recognition module
  - EWC penalty

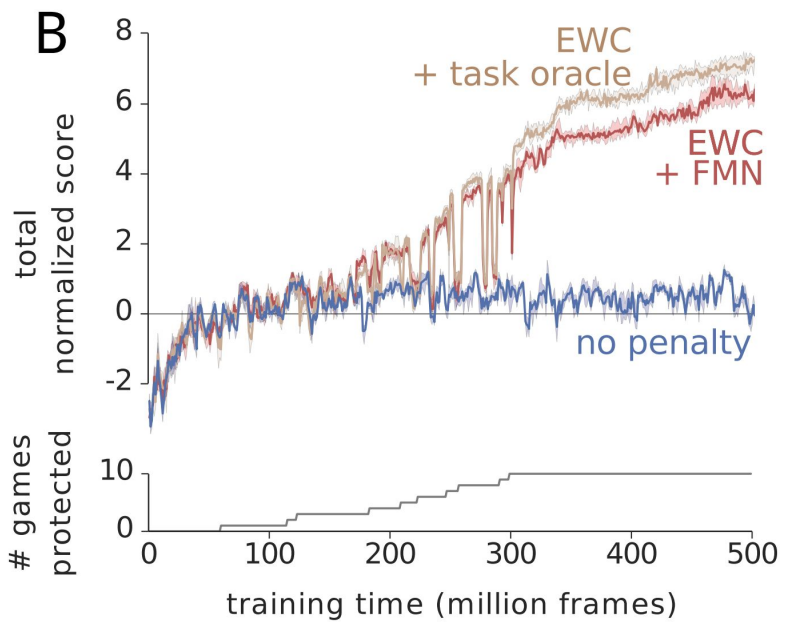
Experiments 2: RL

- Two important modifications compared to prior work
  - Task-recognition module
  - EWC penalty

- Task-recognition module infers which task the current data belongs to
- Whenever the task is switched, they compute the Fisher information
  - Add penalty only for games for which the model had seen at least 20 Mio frames

Results

- Maximum is 10 (human performance on all games)
- Without EWC model can play at most one game well
- With EWC, model learns to play multiple games well



Summary and discussion

- Most important takeaway: We can minimize catastrophic forgetting by constraining the learning process
- They derive a clever way of achieving that and demonstrate empirical succeed on two problem settings: supervised classification (MNIST) , RL (Atari games)

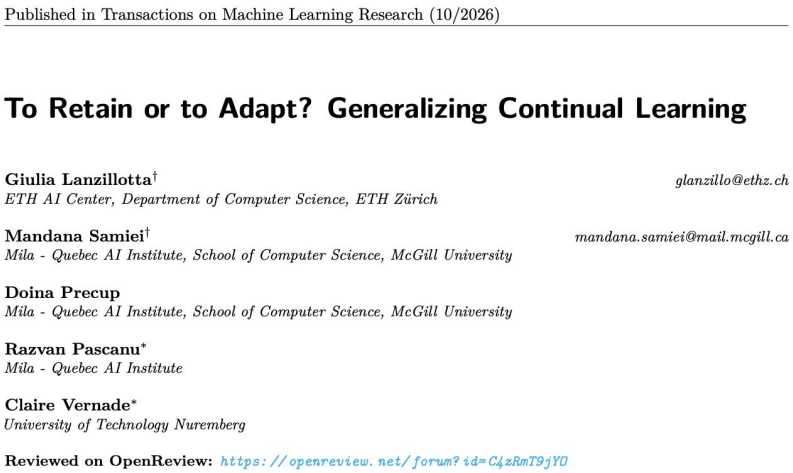
What are the limitations of this approach?

Thank you!

Next lecture: 04.11.2026

Next paper: To Retain or to Adapt? Generalizing Continual Learning

- Presented by Jaleel Ahmed Radhu Khwaja



Published in Transactions on Machine Learning Research (10/2026)

To Retain or to Adapt? Generalizing Continual Learning

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Reviewed on OpenReview: *https://openreview.net/forum?id=C4zRmT9jYO*

Backup slides

Why positive semi-definite matters

$$v^\top F\, v \;=\; \mathbb{E}\Big[\big(v^\top \nabla_\theta \log p(y \mid x, \theta)\big)^2\Big] \;\geq\; 0 \quad \text{for every direction } v$$

The penalty is a bowl
Never negative, and zero only at θ^*_A.

The Gaussian is valid
A covariance needs non-negative variances.

The Hessian can fail
At a saddle, or away from a minimum, it has negative curvature. The Fisher cannot.

The Fisher is an average of outer products, and each one is non-negative in every direction. So the average is too.

Reminder

The Fisher information

$$F_i = \mathbb{E}_{x \sim \mathcal{D}_A}\, \mathbb{E}_{y \sim p(y|x,\theta^*_A)} \left[\left(\frac{\partial \log p(y \mid x, \theta)}{\partial \theta_i} \right)^2 \right]_{\theta = \theta^*_A}$$